NETWORK SECURITY – GROUP ASSIGNMENT

Group 6: Network Security Monitoring and Security Analytics

From Network Traffic to Security Intelligence

Group Size: 3 students
Duration: 6 weeks
Final Presentation: Approximately 30 minutes
Assignment Weight: 30% of the final course grade

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1. Project Description

Modern networks generate a large amount of security-relevant information, including packets, network flows, DNS queries, connection records, authentication events, and system logs. Security monitoring allows organizations to collect and analyze this information to identify suspicious activities and potential security incidents.

However, collecting network data alone does not provide security. Security analysts must determine what information is relevant, identify abnormal behavior, correlate different sources of evidence, and distinguish legitimate activities from potential attacks.

In this project, your group will investigate **Network Security Monitoring (NSM) and Security Analytics**, with a focus on how network data can be collected, analyzed, and transformed into actionable security information.

The project should address the following central question:

How can network traffic and security data be analyzed to identify suspicious activities and support effective security monitoring and incident detection?

The project must combine:

- Network traffic analysis
- Security monitoring concepts
- Packet/flow analysis
- Log analysis
- Threat detection
- Practical investigation

- Security analytics
- Critical evaluation

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2. Project Requirements

A. Network Security Monitoring Fundamentals

Explain and analyze:

- Network Security Monitoring (NSM)
- Network traffic monitoring
- Packet capture
- Network flows
- Metadata
- Logs
- Security events
- Indicators of Compromise (IoCs)
- Indicators of Attack (IoAs)
- Security alerts
- Security analytics

The group should explain the difference between:

Collecting network data

and

Using network data to identify security threats.

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3. Network Data Sources

Investigate at least **three different sources of security information**, such as:

- Packet captures
- DNS traffic

- HTTP/HTTPS traffic
- TCP/UDP flows
- Firewall logs
- Authentication logs
- IDS/IPS alerts
- Network device logs
- Endpoint logs

For each source, discuss:

- What information it provides
- What security events can be identified
- Advantages
- Limitations
- Potential privacy concerns

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4. Practical Network Traffic Analysis

The group must perform a practical investigation using real or generated network traffic in an authorized environment.

Possible tools include:

- Wireshark
- Zeek
- Suricata
- Security Onion
- tcpdump
- ELK / Elastic Stack
- OpenSearch
- Other instructor-approved tools

The group should collect or analyze network data and identify at least **three security-relevant activities or behaviors**.

Examples include:

- Network scanning
- Suspicious DNS activity
- Repeated connection attempts
- Unusual communication patterns
- Suspicious HTTP requests
- Brute-force behavior
- Unexpected outbound connections
- Possible command-and-control behavior
- Abnormal traffic volume

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5. Required Practical Investigation

The group must perform at least **two practical investigations**.

Possible investigations include:

Option A – Packet Analysis

Use Wireshark or another packet-analysis tool to investigate network traffic.

The group should identify:

- Source and destination
- Protocol
- Ports
- Communication pattern
- Relevant packet fields
- Security indicators

Option B – Network Flow Analysis

Analyze network flows to identify unusual communication patterns.

Investigate:

- Source/destination pairs
- Connection frequency
- Duration
- Volume
- Ports
- Protocols
- Temporal patterns

Option C – DNS Security Analysis

Analyze DNS activity to identify suspicious behavior.

Possible indicators include:

- Unusual query frequency
- Unusual domains
- Repeated failed queries
- Long or unusual domain names
- Unexpected external destinations

Option D – Log Correlation

Combine at least two sources of security information, such as:

Firewall Logs + DNS Logs

or:

IDS Alerts + Packet Captures

The objective is to determine whether multiple pieces of evidence support the same security event.

Option E – Security Dashboard

Develop or configure a basic monitoring dashboard using an appropriate platform.

The dashboard may display:

- Traffic volume
- Top connections
- Top source/destination addresses
- Protocol distribution
- Alerts
- Suspicious activities
- Time-based activity

The dashboard should support security analysis rather than simply display statistics.

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6. Security Investigation Scenario

The group should develop at least **one realistic security investigation scenario**.

For example:

A workstation inside an organization is generating unusual outbound traffic. The security team needs to determine whether the activity represents normal behavior or a potential security incident.

The group should investigate:

1. Initial security indicator
2. Available network evidence
3. Traffic analysis
4. Related logs
5. Suspicious behavior

6. Possible explanation
7. Additional evidence required
8. Security conclusion
9. Recommended response

The group should explain how a security analyst would proceed from an initial alert to a reasoned conclusion.

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7. Indicators of Compromise

The project must discuss Indicators of Compromise (IoCs).

Possible IoCs include:

- IP addresses
- Domain names
- URLs
- Hashes
- Unusual ports
- Suspicious DNS requests
- Abnormal network connections
- Repeated authentication failures
- Unexpected outbound communication

The group should explain:

- How IoCs can be identified
- How IoCs can be used in network monitoring
- Limitations of IoC-based detection
- Why behavioral analysis may be necessary

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8. Detection and Security Analytics

The group should investigate how security analysts identify suspicious behavior.

Possible approaches include:

- Signature-based detection
- Rule-based detection
- Threshold-based detection
- Statistical analysis
- Behavioral analysis
- Anomaly detection
- Correlation
- Threat intelligence

The group should explain the strengths and limitations of different approaches.

In particular, discuss the difference between:

Known indicators

and

Unknown or previously unseen behavior.

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9. Analysis of False Positives and False Negatives

The group should discuss the problem of incorrect security alerts.

False Positive

Normal activity is incorrectly identified as suspicious.

False Negative

Suspicious or malicious activity is not identified.

The group should discuss:

- Causes
- Security impact
- Analyst workload

- Alert fatigue
- Rule tuning
- Baseline creation
- Detection thresholds

Where possible, the practical experiment should include an example of an activity that could reasonably generate a false positive or ambiguous alert.

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10. Security Monitoring Architecture

Design a network security monitoring architecture for a hypothetical organization.

The proposed architecture should show:

- Network traffic sources
- Sensors
- Packet/flow collection
- Log collection
- Analysis engine
- Security alerts
- Dashboard
- Security analyst
- Incident-response process

Possible components include:

- Network sensors
- IDS/IPS
- Firewall
- DNS monitoring
- SIEM
- Packet capture

- Network flow monitoring
- Security analytics

The group should explain how information flows through the architecture.

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11. Final Report

Submit a **15–20 page report**, excluding references and appendices.

The report should contain:

1. **Introduction and Project Objectives**
2. **Network Security Monitoring Fundamentals**
3. **Network Data Sources**
4. **Security Analytics and Detection**
5. **Related Work / Case Study**
6. **Laboratory Environment**
7. **Experimental Methodology**
8. **Network Traffic / Log Analysis**
9. **Security Investigation Scenario**
10. **Experimental Results**
11. **Security Analysis and Discussion**
12. **Proposed Monitoring Architecture**
13. **Recommendations**
14. **Conclusion**
15. **References**
16. **Appendices**, if necessary

The report should include appropriate:

- Network diagrams
- Packet captures

- Log samples
- Tables
- Traffic statistics
- Screenshots
- Security dashboards
- Investigation evidence

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12. Final Presentation

Each group will give a presentation of approximately **30 minutes**.

The presentation should include:

- Network monitoring problem
- NSM concepts
- Data sources
- Practical investigation
- Network traffic analysis
- Security indicators
- Investigation scenario
- Results
- Security monitoring architecture
- Recommendations
- Conclusion

The group should demonstrate its analysis tools or monitoring environment where feasible.

Each member must participate in the presentation.

The instructor may ask any member questions about any part of the project.

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13. Deliverables

At the end of Week 6, submit:

1. Final Report

PDF format.

2. Presentation Slides

PowerPoint or PDF format.

3. Practical Materials

Where applicable:

- Packet captures
- Network logs
- Analysis scripts
- Detection rules
- Dashboards
- Screenshots
- Experimental data
- Network diagrams

4. Individual Contribution Statement

Each student must briefly describe their contribution.

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14. Assessment and Grading

The assignment is worth **30% of the final course grade**.

Assessment Component	Points
NSM Technical Research and Understanding	15
Network Data and Traffic Analysis	15
Practical Investigation	20
Security Evidence and IoC Analysis	10

Assessment Component	Points
Security Analytics and Detection	10
Investigation Scenario and Critical Analysis	10
Monitoring Architecture	5
Final Report	5
Presentation and Demonstration	5
Q&A / Individual Technical Understanding	5
Total	**100**

The resulting assignment score will contribute **30% to the overall course grade**.

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15. Assessment Criteria

Excellent (90–100)

The project demonstrates strong understanding of network security monitoring, meaningful analysis of network data, well-designed investigations, reliable security evidence, and technically justified conclusions.

Good (75–89)

The project demonstrates good technical understanding and practical analysis, with appropriate evidence and recommendations, but some areas may lack depth.

Satisfactory (60–74)

The project addresses the major requirements but provides limited investigation, evidence, or critical analysis.

Below Expectations (<60)

The project is incomplete, primarily descriptive, lacks meaningful network-data analysis, or demonstrates limited understanding of security monitoring.

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16. Security and Academic Requirements

All network monitoring and traffic-capture activities must be performed in an **authorized environment**.

Students must not capture, inspect, or analyze communications belonging to other users without explicit authorization.

The practical environment should preferably use:

- Virtual machines
- Containers
- Laboratory networks
- Publicly available datasets
- Instructor-approved traffic captures

Students should take appropriate care when handling potentially sensitive network data.

All experimental results must be produced or legitimately obtained by the group. Fabricated results or references are not acceptable.

Generative AI tools may be used as supporting tools, but students are responsible for verifying technical information, references, and experimental results. Any use of AI should be disclosed briefly in the report.

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17. Key Expectation

The project should demonstrate the complete security-monitoring process:

Network Data → Collection → Analysis → Security Indicator → Investigation → Detection → Decision → Response

The group should be able to answer:

What network data should a security analyst collect?

What information can be extracted from packets, flows, and logs?

How can normal and suspicious behavior be distinguished?

How can multiple pieces of evidence be correlated?

What are the limitations of IoC-based detection?

How can security analytics identify suspicious behavior?

How should an organization design a practical network security monitoring architecture?

The main objective is to demonstrate that **network security monitoring is not simply the collection of packets and logs. Its value comes from transforming network data into reliable security evidence that can support detection, investigation, and incident response.**